

RFQ Portal — Complete Project File Inventory & Architecture Directory

Project Name: RFQ Portal (Government of Andhra Pradesh / AGIC)

Repository: <https://github.com/kirantejatj/rfq-portal.git>

Root Directory: `C:\Users\AppData\gemini\antigravity\scratch\rfq-portal`

Stack: Python FastAPI, PostgreSQL (SQLAlchemy + Pydantic), React (Vite + Tailwind CSS + Lucide Icons)

Table of Contents

- [1. Root System Files & Automation Scripts](#)
- [2. Backend Architecture & Files \(`backend/` \)](#)
 - [Core Configuration & Security \(`backend/app/core/` \)](#)
 - [Database Models & ORM Schema \(`backend/app/models/` \)](#)
 - [Pydantic Request/Response Schemas \(`backend/app/schemas/` \)](#)
 - [API Routers & Business Logic \(`backend/app/routers/` \)](#)
 - [Entrypoint & Database Seeding \(`backend/` \)](#)
- [3. Frontend Architecture & Files \(`frontend/` \)](#)
 - [Configuration & Build Setup \(`frontend/` \)](#)
 - [Core API Client & Authentication State \(`frontend/src/` \)](#)
 - [Reusable UI Components \(`frontend/src/components/` \)](#)
 - [Public & Authentication Pages \(`frontend/src/pages/auth/` \)](#)
 - [Vendor Portal Pages \(`frontend/src/pages/applicant/` \)](#)
 - [Officer Portal Pages \(`frontend/src/pages/ce/` \)](#)
 - [Developer API Workbench \(`frontend/src/pages/admin/` \)](#)
- [4. Master Documentation & Knowledge Transfer Artifacts](#)
- [5. Database Schema & Table Summary](#)

1. Root System Files & Automation Scripts

File Path	Purpose & What It Contains
START_PORTAL.bat	One-Click Startup Script: Batch script that checks and starts the PostgreSQL database, launches the FastAPI backend on port <code>8000</code> , and launches the Vite React frontend on port <code>5173</code> . Opens the browser automatically.
STOP_PORTAL.bat	One-Click Shutdown Script: Gracefully terminates background Python (uvicorn) and Node.js (Vite) processes running the portal.
SYSTEM DESIGN AND DB DOCUMENTATION.md	Architecture Specification: Complete markdown document outlining PostgreSQL table structures, foreign keys, lifecycle states, authentication mechanisms, and API design.
AGIC_RFQ_Portal_Complete_Master_Documentation.html	Master HTML Documentation: Beautiful offline standalone HTML reference manual including schema diagrams, sequence flows, workflows, and system architecture.
AGIC_RFQ_Portal_KT_and_User_Guide.html	Knowledge Transfer & User Manual: Interactive HTML user guide for Vendors and Officers with role-based walkthroughs, step-by-step submission instructions, and FAQs.
RFQ_Portal_Design_and_Architecture_Documentation.html	Design & Architecture Visual Guide: Formatted HTML document covering engineering designs, component interactions, and state machines.

2. Backend Architecture & Files (`backend/`)

2.1 Core Configuration & Security

Located at: `backend/app/core/`

File Path	Purpose & What It Contains
backend/app/core/config.py	Environment & App Settings: Pydantic BaseSettings class loading database connection strings (<code>DATABASE_URL</code>), JWT secret key (<code>SECRET_KEY</code>), algorithm (<code>HS256</code>), token expiration time, upload folder paths (<code>UPLOAD_DIR</code>), and project metadata.
backend/app/core/database.py	Database Connection Engine: Initializes SQLAlchemy engine for PostgreSQL, configures <code>sessionmaker</code> (<code>SessionLocal</code>), sets up declarative <code>Base</code> , and exports the <code>get_db</code> dependency for route injection.
backend/app/core/security.py	Cryptography & Token Generation: Password hashing and verification using <code>passlib.context.CryptContext</code> with <code>bcrypt</code> . Generates signed JWT bearer tokens (<code>create_access_token</code>) with role, subject, and expiry timestamps.
backend/app/core/dependencies.py	RBAC Auth Guards & Middleware: Dependency injectors: <code>get_current_user</code> (decodes JWT and loads user model), <code>require_ce</code> (restricts access to Chief Engineers/Officers), <code>require_applicant</code> (restricts access to Vendors), and <code>require_super_admin</code> .

2.2 Database Models & ORM Schema

Located at: `backend/app/models/`

File Path	Purpose & What It Contains
backend/app/models/schema.py	SQLAlchemy Database Models: Defines all relational tables: • <code>CEUser</code> (Officer accounts with designation & credentials) • <code>Applicant</code> (Vendor profiles with <code>vendor_type</code>, GSTIN, PAN, turnover, experience) • <code>ApplicantDocument</code> (Vendor uploaded certificates) • <code>OTPVerification</code> (One-time password audit records) • <code>RFQTender</code> (RFQ definitions with <code>quotation_valid_upto</code>, <code>validity_period</code>, dates, budgets, creator ID) • <code>RFQJob</code> (Granular RFQ Items with <code>category</code> ['Supply Item'/'Only Rate'], <code>unit_rate</code>, <code>estimated_quantity</code>, <code>amount</code>) • <code>TenderDocument</code> (Official attachments/specifications) • <code>Application</code> (Vendor quotation submissions with <code>quoted_amount</code>, covering letter, status) • <code>ApplicationJob</code> (Vendor itemized selections and custom amounts) • <code>TechnicalFinancialCapability</code> (Annexure II past works) • <code>ApplicationProposalItem</code> (Annexure III line item price breakup) • <code>EMDPayment</code> (Earnest Money Deposit references) • <code>ApplicationDocument</code> (Quotation attachments) • <code>ApplicationStatusHistory</code> (Audit trail of status changes) • <code>Clarification</code> (Pre-bid vendor queries and officer replies).

2.3 Pydantic Request/Response Schemas

Located at: `backend/app/schemas/`

File Path	Purpose & What It Contains
backend/app/schemas/auth.py	Auth Payloads: Pydantic schemas for OTP generation (<code>SendOTPRequest</code>), verification (<code>VerifyOTPRequest</code>), vendor registration (<code>ApplicantRegisterRequest</code> with <code>vendor_type</code>), vendor login (<code>ApplicantLoginRequest</code>), officer login (<code>CELoginRequest</code>), and JWT token response (<code>TokenResponse</code>).
backend/app/schemas/tender.py	RFQ & Items Payloads: Validation models for RFQ Item creation/output (<code>RFQJobCreate</code>, <code>RFQJobOut</code> with <code>category</code>, <code>unit_rate</code>, <code>amount</code>), RFQ creation/update (<code>TenderCreate</code>, <code>TenderUpdate</code> with <code>quotation_valid_upto</code>, <code>validity_period</code>), and full RFQ response (<code>TenderOut</code>).
backend/app/schemas/application.py	Quotation Submission Payloads: Validation models for quotation submission (<code>ApplicationSubmitRequest</code>), status change (<code>ApplicationStatusUpdate</code>), Annexure II (<code>CapabilityItemSchema</code>), Annexure III price breakup (<code>ProposalItemSchema</code>), EMD details (<code>EMDPaymentSchema</code>), and comprehensive response model (<code>ApplicationOut</code>).
backend/app/schemas/clarification.py	Clarification Payloads: Models for asking pre-bid queries (<code>ClarificationCreate</code>), answering queries (<code>ClarificationReply</code>), and public clarification list (<code>ClarificationOut</code>).

2.4 API Routers & Business Logic

Located at: `backend/app/routers/`

File Path	Purpose & What It Contains
<code>backend/app/routers/auth.py</code>	Authentication Endpoints: • <code>POST /api/auth/applicant/request-otp</code> • <code>POST /api/auth/applicant/verify-otp</code> • <code>POST /api/auth/applicant/register</code> (Stores <code>vendor_type</code> & marks eligibility) • <code>POST /api/auth/applicant/login</code> • <code>POST /api/auth/ce/login</code> (Officer authentication) • <code>GET /api/auth/me</code> (Returns current user profile + <code>is_eligible</code> calculation) • <code>POST /api/auth/applicant/{id}/documents</code> (Profile document uploads).
<code>backend/app/routers/tenders.py</code>	RFQ Lifecycle Endpoints: • <code>GET /api/tenders</code> (List published/active RFQs with search & category filters) • <code>GET /api/tenders/{id}</code> (Detailed view of RFQ + RFQ Items + validity dates) • <code>POST /api/tenders</code> (Officer creates new RFQ with RFQ Items and validity period) • <code>PUT /api/tenders/{id}</code> (Update RFQ metadata) • <code>POST /api/tenders/{id}/jobs</code> (Add RFQ Item) • <code>DELETE /api/tenders/{id}/jobs/{job_id}</code> (Delete RFQ Item) • <code>PATCH /api/tenders/{id}/status</code> (Publish, extend, or close RFQ) • <code>POST /api/tenders/{id}/documents</code> (Attach official documents).
<code>backend/app/routers/applications.py</code>	Quotation Submissions & Officer Review: • <code>POST /api/applications</code> (Vendor submits quotation: validates vendor eligibility, saves custom amounts, Annexures II & III, and EMD) • <code>GET /api/applications/my</code> (Vendor views own submitted quotations) • <code>GET /api/applications/tender/{id}</code> (Officer Authorization Guard: Enforces that only the creator Officer can view submissions; returns <code>403 Forbidden</code> to other officers) • <code>GET /api/applications/{id}</code> (Detailed quotation dossier view with role checks) • <code>PATCH /api/applications/{id}/status</code> (Officer Evaluation Decision: Creator Officer accepts/rejects quotation and logs audit trail) • <code>POST /api/applications/{id}/documents</code> (Upload submission attachments) • <code>GET /api/applications/{id}/download-all</code> (Compiles all vendor and quotation documents into a single dynamic ZIP archive).
<code>backend/app/routers/clarifications.py</code>	Clarifications / Pre-Bid Queries: • <code>GET /api/clarifications/tender/{id}</code> (List public Q&A for an RFQ) • <code>POST /api/clarifications</code> (Vendor submits a query) • <code>POST /api/clarifications/{id}/reply</code> (Officer posts official reply).
<code>backend/app/routers/stats.py</code>	Analytics & Scoped Dashboard Metrics: • <code>GET /api/stats/ce</code> (Calculates RFQs raised, total quotations received, pending reviews, and approved awards strictly scoped to the logged-in Officer) • <code>GET /api/stats/applicant</code> (Calculates quotations submitted, under review, and awards for the vendor).

2.5 Entrypoint & Database Seeding

Located at: `backend/`

File Path	Purpose & What It Contains
<code>backend/app/main.py</code>	FastAPI Application Entry: Configures CORS middleware, mounts <code>/uploads</code> static file server, registers all routers, customizes OpenAPI metadata, and mounts Scalar API reference documentation (<code>/scalar</code>).
<code>backend/run.py</code>	Uvicorn Server Launcher: Boots <code>app.main:app</code> on host <code>0.0.0.0</code> , port <code>8000</code> with auto-reloading enabled.
<code>backend/seed.py</code>	Database Seeding Utility: Pre-populates PostgreSQL with initial Officers, sample eligible Vendors, active RFQs with RFQ Items, and submitted quotation records for instant demo/testing.

3. Frontend Architecture & Files (`frontend/`)

3.1 Configuration & Build Setup

Located at: `frontend/`

File Path	Purpose & What It Contains
<code>frontend/package.json</code>	NPM Package Manifest: Defines project dependencies (React 18, React Router v6, Axios, Lucide-React, TailwindCSS) and scripts (<code>dev</code> , <code>build</code> , <code>preview</code>).
<code>frontend/vite.config.js</code>	Vite Build Configuration: Configures React plugin, development server port (<code>5173</code>), and build bundling targets.
<code>frontend/tailwind.config.js</code>	Tailwind Design Theme: Defines government-themed color palettes (<code>gov-50</code> to <code>gov-900</code> , gold accents, font families, and container layouts).
<code>frontend/postcss.config.js</code>	PostCSS Pipeline: Connects TailwindCSS and Autoprefixer to Vite CSS transformation engine.
<code>frontend/index.html</code>	Single Page HTML Host: Root HTML file loading Google Fonts (Inter, Outfit), favicon, and mounting <code><div id="root"></code> .

3.2 Core API Client & Authentication State

Located at: `frontend/src/`

File Path	Purpose & What It Contains
<code>frontend/src/main.jsx</code>	React DOM Root: Bootstraps the React tree, wraps in <code>BrowserRouter</code> , and mounts into DOM.
<code>frontend/src/App.jsx</code>	Application Router & Layout: Defines all application routes, role-based route guards (<code>PrivateRoute</code>), global notification banners, and navigation wrappers (<code>Navbar</code> , <code>Footer</code>).

File Path	Purpose & What It Contains
frontend/src/api/client.js	Axios HTTP Client: Centralized HTTP client configured with baseURL (<code>http://localhost:8000/api</code>). Attaches Bearer JWT from <code>localStorage</code> on every request and handles global 401/403 errors.
frontend/src/context/AuthContext.jsx	Authentication Context Provider: Global React Context managing user authentication state, current user profile (<code>user</code>), active role (<code>APPLICANT</code> , <code>CE</code> , <code>SUPER_ADMIN</code>), login, logout, and token persistence.
frontend/src/index.css & frontend/src/App.css	Global Styles: Tailwind layer imports, custom scrollbar styling, glassmorphic styles, print stylesheets, and animation keyframes.

3.3 Reusable UI Components

Located at: `frontend/src/components/`

File Path	Purpose & What It Contains
frontend/src/components/Navbar.jsx	Primary Navigation Header: Government portal branding (RFQ Portal), active RFQ browse link, contextual role-based buttons (Vendor Workspace, Officer Panel), user profile avatar, and logout trigger. Hidden developer workbench link.
frontend/src/components/Footer.jsx	Standard Government Footer: Official AGIC / Andhra Pradesh government notices, quick navigation links, helpline contacts, and copyright information.
frontend/src/components/StatusBadge.jsx	Visual Status Pill Component: Dynamic color-coded status badges for RFQs (<code>PUBLISHED</code> , <code>CLOSED</code> , <code>DRAFT</code>) and Quotations (<code>SUBMITTED</code> , <code>UNDER_REVIEW</code> , <code>ACCEPTED</code> , <code>REJECTED</code> , <code>WITHDRAWN</code>).
frontend/src/components/CountdownTimer.jsx	Live Submission Countdown: Real-time timer showing Days, Hours, Minutes, and Seconds remaining until quotation deadline. Turns red when closing soon.

3.4 Public & Authentication Pages

Located at: `frontend/src/pages/` and `frontend/src/pages/auth/`

File Path	Purpose & What It Contains
frontend/src/pages/Home.jsx	Public Portal Landing Page: Hero section with official state emblem, live stats counter (Active RFQs, Value in Crores, Registered Vendors), live search bar, category filters, and active RFQ cards.
frontend/src/pages/TenderDetails.jsx	RFQ Detailed Dossier Page: Full public specification view: EMD requirements, Quotation Validity Period , RFQ Items Schedule (with <code>Supply Item</code> / <code>Only Rate</code> categories, unit rates, quantities), pre-bid Q&A thread, and "Quote for this RFQ" button.
frontend/src/pages/auth/ApplicantLogin.jsx	Vendor Portal Login: OTP-based and password-based login for Vendors. Quick test login credentials for one-click developer testing.

File Path	Purpose & What It Contains
frontend/src/pages/auth/ApplicantRegister.jsx	Vendor Registration Page: Registration form enforcing selection of the 4 eligible vendor categories (Manufacturer , Authorised Dealer , Authorised Distributor , Contractor), mobile OTP verification, GSTIN, PAN, and credentials.
frontend/src/pages/auth/CELogin.jsx	Officer Portal Login: Secure departmental login for Chief Engineers and Executive Officers with designation verification.

3.5 Vendor Portal Pages

Located at: [frontend/src/pages/applicant/](#)

File Path	Purpose & What It Contains
frontend/src/pages/applicant/ApplicantDashboard.jsx	Vendor Workspace Hub: Displays Vendor Category badge, Eligibility status badge, submission metrics, quick links to active RFQs, and recent quotation statuses.
frontend/src/pages/applicant/MyApplications.jsx	Submitted Quotations Matrix: Tabular view of all quotations submitted by the logged-in vendor, custom quoted amounts (₹), submission timestamps, and current review states.
frontend/src/pages/applicant/SubmitQuotation.jsx	Multi-Step Quotation Submission Wizard: • Eligibility Guard: Warns/blocks ineligible vendors. • Step 1: Covering Letter: Authorised signatory details & dates. • Step 2: RFQ Items Selection: Line-by-line item selection with unit rate and auto-calculated amounts. • Step 3: Annexure II: Past technical & financial works declarations. • Step 4: Annexure III: Commercial proposal, itemized pricing breakdown, tax inclusion, and quotation validity affirmations. • Step 5: EMD & Documents: EMD transaction reference and PDF document uploads.

3.6 Officer Portal Pages

Located at: [frontend/src/pages/ce/](#)

File Path	Purpose & What It Contains
frontend/src/pages/ce/CEDashboard.jsx	Officer Executive Panel: Summary of RFQs raised by the logged-in officer, quotation response counts, pending technical reviews, action shortcuts, and RFQ management table.
frontend/src/pages/ce/CreateTender.jsx	Raise New RFQ Form: Interface for Officers to draft and publish RFQs: sets scope of work, budget, EMD, Quotation Validity Period & Date , and dynamically adds multiple RFQ Items (Supply Item vs Only Rate , Unit Rate, Quantity, Amount).

File Path	Purpose & What It Contains
frontend/src/pages/ce/TenderSubmissions.jsx	Quotation Submissions Matrix: • Officer Authorization Guard: Validates that only the Officer who raised this RFQ can access this page. • Comparative evaluation table of all vendor quotations sorted by quoted price. • Shows Vendor Category, Quoted Amount (₹), Submission Date, and Evaluation trigger.
frontend/src/pages/ce/ApplicationReview.jsx	Quotation Dossier & Evaluation Panel: • Officer Authorization Guard: 403 Forbidden protection. • Full vendor profile (Category, Turnover, Experience, PAN/GSTIN). • Line item pricing schedule breakdown. • Download All Documents ZIP button. • Officer Evaluation Decision Box: Change status (ACCEPTED (L1 Award) , REJECTED , UNDER_REVIEW), record audit remarks, and update database.

3.7 Developer API Workbench

Located at: [frontend/src/pages/admin/](#)

File Path	Purpose & What It Contains
frontend/src/pages/admin/ApiWorkbench.jsx	Interactive Developer API Workbench: Standalone developer console accessible directly at /developer and /api-explorer . Allows developers to inspect, test, and execute all backend REST endpoints directly with quick-login JWT tokens and JSON payload editors.

4. Master Documentation & Knowledge Transfer Artifacts

Document File	Format	Target Audience & Contents
SYSTEM DESIGN AND DB DOCUMENTATION.md	Markdown	Developers & DBAs: Database schema, DDL definitions, indexing strategies, entity relationships, and security protocols.
AGIC RFQ Portal Complete Master Documentation.html	Standalone HTML	Technical Leads & Stakeholders: Complete architectural breakdown, sequence diagrams, API reference, and deployment guide.
AGIC RFQ Portal KT and User Guide.html	Standalone HTML	Vendors & Department Officers: Step-by-step Knowledge Transfer (KT) guide with visual flowcharts, login guides, and submission walkthroughs.
RFQ Portal Design and Architecture Documentation.html	Standalone HTML	Executive Overview: High-level design document and engineering specifications.

5. Database Schema & Table Summary

Table Name	Primary Key	Key Columns & Foreign Keys	Description
ce_users	ce_id	name, mobile_no, email, password_hash, designation, role, is_active	Departmental Officers (Chief Engineers, Executive Engineers)
applicants	applicant_id	firm_name, mobile_no, vendor_type, registration_type, gstin, pan_no, turnover, work_experience	Registered Vendors (Manufacturers, Dealers, Distributors, Contractors)
applicant_documents	document_id	applicant_id (FK), document_type, file_name, file_path, file_size_kb	Vendor profile attachments (Turnover certificate, PAN card)
otp_verifications	otp_id	mobile_no, otp_code, purpose, is_verified, expires_at	OTP logs for mobile authentication
rfq_tenders	tender_id	tender_ref_no, title, created_by (FK $\rightarrow$ ce_users), quotation_valid_upto, validity_period, status, emd_amount	RFQ records created and managed by Officers
rfq_jobs	job_id	tender_id (FK), job_name, category (Supply Item / Only Rate), unit_rate, estimated_quantity, amount	Partitioned RFQ Items under an RFQ
tender_documents	tender_document_id	tender_id (FK), document_type, file_name, file_path	Official tender documents and technical specs
applications	application_id	tender_id (FK), applicant_id (FK), application_no, quoted_amount, status, remarks	Vendor quotation submissions
application_jobs	application_job_id	application_id (FK), job_id (FK), quoted_amount, remarks, status	Vendor-selected RFQ Items & quoted rates
application_proposal_items	proposal_item_id	application_id (FK), job_id (FK), item_description, quantity, unit, rate_per_unit, amount	Annexure III line-by-line financial breakup

Table Name	Primary Key	Key Columns & Foreign Keys	Description
technical_financial_capabilities	capability_id	application_id (FK), sl_no , work_description , client_name , cost_lakhs	Annexure II past work experience declarations
emd_payments	emd_id	application_id (FK), amount , payment_mode , transaction_ref	Earnest Money Deposit payment records
application_status_history	history_id	application_id (FK), old_status , new_status , changed_by_ce (FK), remarks	Audit trail of all quotation status updates
clarifications	clarification_id	tender_id (FK), applicant_id (FK), query_text , reply_text , status	Pre-bid clarifications and officer responses